

Ch 12: Magnetic Effects of Electric Current

Question Bank | 2M | 3M | 5M

SECTION A - 2 MARKS

Q1. What is meant by the magnetic effect of electric current? Name the scientist who first discovered it. (PYQ Pattern)

Q2. What is a magnetic field? Write the SI unit of magnetic field. (PYQ Pattern)

Q3. What are magnetic field lines? State any two properties of magnetic field lines. (PYQ Pattern)

Q4. Why do magnetic field lines never intersect each other? Give reason. (PYQ Pattern)

Q5. How can the direction of magnetic field at a point be found using a compass needle? (PYQ Pattern)

Q6. Draw the magnetic field pattern around a bar magnet. Mark the direction of field lines. (PYQ Pattern)

Q7. What is the direction of magnetic field lines outside a bar magnet and inside a bar magnet? (PYQ Pattern)

Q8. What happens to the compass needle when it is brought near a current-carrying conductor? Why? (PYQ Pattern)

Q9. State the right-hand thumb rule. What does the thumb represent in this rule? (PYQ Pattern)

Q10. Draw the magnetic field lines around a straight current-carrying conductor. (PYQ Pattern)

Q11. How does the strength of magnetic field around a straight conductor change with distance from the conductor? (PYQ Pattern)

Q12. What happens to the magnetic field around a conductor if the current through it is increased? (PYQ Pattern)

Q13. What is the shape of magnetic field lines around a straight current-carrying wire? (PYQ Pattern)

Q14. Draw the magnetic field pattern due to a circular current-carrying loop. (PYQ Pattern)

Q15. What is the nature of magnetic field at the centre of a circular current-carrying loop? (PYQ Pattern)

Q16. State two factors on which the magnetic field at the centre of a circular loop depends. (PYQ Pattern)

Q17. What is a solenoid? What type of magnetic field is produced inside a current-carrying solenoid? (PYQ Pattern)

Q18. How can a current-carrying solenoid behave like a bar magnet? (PYQ Pattern)

Q19. What is an electromagnet? Name two materials commonly used to make the core of an electromagnet. (PYQ Pattern)

Q20. Write two differences between a permanent magnet and an electromagnet. (PYQ Pattern)

Q21. What is meant by the force on a current-carrying conductor placed in a magnetic field? (PYQ Pattern)

Q22. State Fleming's left-hand rule. Which finger represents the direction of force? (PYQ Pattern)

Q23. Name the rule used to find the direction of force on a current-carrying conductor in a magnetic field. (PYQ Pattern)

Q24. When is the force on a current-carrying conductor in a magnetic field maximum? (PYQ Pattern)

Q25. When is the force on a current-carrying conductor in a magnetic field zero? (PYQ Pattern)

Q26. What is an electric motor? State the energy conversion taking place in an electric motor. (PYQ Pattern)

Q27. Name the device used to reverse the direction of current in the coil of a DC motor. (PYQ Pattern)

Q28. What is the function of the split-ring commutator in an electric motor? (PYQ Pattern)

Q29. What is electromagnetic induction? Name the scientist associated with this discovery. (PYQ Pattern)

Q30. State Fleming's right-hand rule. What does the forefinger represent? (PYQ Pattern)

Q31. What is an electric generator? State the energy conversion in an electric generator. (PYQ Pattern)

Q32. Write one difference between an electric motor and an electric generator. (PYQ Pattern)

Q33. What is alternating current? What is direct current? Give one difference. (PYQ Pattern)

Q34. What is the frequency of AC supply in India? How many times does it change direction in one second? (PYQ Pattern)

Q35. What is a domestic electric circuit? Name the three wires used in household circuits. (PYQ Pattern)

Q36. What are live wire, neutral wire and earth wire? Mention their usual colour codes. (PYQ Pattern)

Q37. What is earthing? Why is earthing necessary in domestic appliances? (PYQ Pattern)

Q38. What is a fuse? Why is it connected in series with the live wire? (PYQ Pattern)

Q39. What is short circuiting? State one cause of short circuiting. (PYQ Pattern)

Q40. What is overloading? State one cause of overloading in domestic circuits. (PYQ Pattern)

SECTION B - 3 MARKS

Q1. Describe Oersted's experiment to show that an electric current produces a magnetic field. What conclusion is drawn from this experiment? (PYQ Pattern)

Q2. Explain magnetic field lines. Write any three properties of magnetic field lines. (PYQ Pattern)

Q3. Draw the magnetic field pattern around a bar magnet and explain why field lines are crowded near the poles. (PYQ Pattern)

Q4. State the right-hand thumb rule and apply it to find the direction of magnetic field when current flows vertically upward in a straight wire. (PYQ Pattern)

Q5. Describe the magnetic field produced around a straight current-carrying conductor. Mention the effect of current and distance. (PYQ Pattern)

Q6. A current flows through a straight conductor from south to north. Using right-hand thumb rule, find the direction of magnetic field at a point east of the conductor. (PYQ Pattern)

Q7. Draw and explain the magnetic field pattern due to a circular current-carrying loop. What is the direction of field at its centre? (PYQ Pattern)

Q8. How does the magnetic field at the centre of a circular loop change when: (i) current is increased (ii) number of turns is increased (iii) radius is increased? (PYQ Pattern)

Q9. What is a solenoid? Draw the magnetic field pattern of a current-carrying solenoid and compare it with that of a bar magnet. (PYQ Pattern)

Q10. Why is the magnetic field inside a solenoid strong and uniform? Explain with a labelled diagram. (PYQ Pattern)

Q11. Explain how an electromagnet is made. State two methods to increase the strength of an electromagnet. (PYQ Pattern)

Q12. Compare permanent magnet and electromagnet on the basis of material, strength, switching on/off and uses. (PYQ Pattern)

Q13. Describe the force experienced by a current-carrying conductor placed in a magnetic field. State the factors affecting the magnitude of force. (PYQ Pattern)

Q14. State Fleming's left-hand rule. Explain how it is used to find the direction of motion of a conductor. (PYQ Pattern)

Q15. A current-carrying conductor is placed perpendicular to a magnetic field. What happens when: (i) current is reversed (ii) magnetic field is reversed (iii) both are reversed? (PYQ Pattern)

Q16. Explain the principle of an electric motor. Name the main parts of a simple DC motor. (PYQ Pattern)

Q17. What is the role of split rings and carbon brushes in an electric motor? (PYQ Pattern)

Q18. Describe electromagnetic induction. Mention two ways by which induced current can be produced in a coil. (PYQ Pattern)

Q19. State Fleming's right-hand rule. How is it different from Fleming's left-hand rule? (PYQ Pattern)

Q20. Explain the principle of an electric generator. Name the main parts of an AC generator. (PYQ Pattern)

Q21. Compare electric motor and electric generator on the basis of principle, energy conversion and rule used. (PYQ Pattern)

Q22. Differentiate between alternating current and direct current. Why is AC preferred for long-distance transmission? (PYQ Pattern)

Q23. Explain the domestic electric circuit. Why are appliances connected in parallel rather than in series? (PYQ Pattern)

Q24. Write the functions of live wire, neutral wire and earth wire in a household circuit. (PYQ Pattern)

Q25. What is the importance of earthing in electrical safety? Explain with the example of a metallic appliance. (PYQ Pattern)

Q26. Explain the working of an electric fuse. Why should the fuse wire have low melting point and suitable current rating? (PYQ Pattern)

Q27. What is the difference between short circuiting and overloading? State one cause and one prevention for each. (PYQ Pattern)

Q28. A house has many high-power appliances connected to the same socket. Explain the risk involved and the protective device used. (PYQ Pattern)

Q29. Answer the following: (i) Why is the earth wire thick? (ii) Why is the switch connected to live wire? (iii) Why is fuse placed in live wire? (PYQ Pattern)

Q30. Draw a labelled diagram of a domestic electric circuit showing live wire, neutral wire, earth wire, fuse and switch. (PYQ Pattern)

Q31. Explain why a magnetic field is a vector quantity. How is its direction represented using field lines? (PYQ Pattern)

Q32. Why does a current-carrying solenoid have a north pole and a south pole? How can the polarity be reversed? (PYQ Pattern)

Q33. Write three uses of electromagnets and explain why electromagnets are preferred in those applications. (PYQ Pattern)

Q34. Explain how a current-carrying conductor can be made to rotate continuously in a magnetic field. (PYQ Pattern)

Q35. A coil is moved quickly towards a magnet. Explain what is observed in a galvanometer and why. (PYQ Pattern)

SECTION C - 5 MARKS

Q1. Describe Oersted's experiment in detail with a labelled diagram. Explain the observations when current is passed, reversed and stopped. State the conclusion about the magnetic effect of current. (PYQ Pattern)

Q2. Write a detailed note on magnetic field lines. Include definition, direction, properties, field pattern around a bar magnet and reason why two field lines cannot intersect. (PYQ Pattern)

Q3. Explain the magnetic field due to a straight current-carrying conductor. Draw the field pattern, state the right-hand thumb rule and discuss the effect of current and distance from the wire. (PYQ Pattern)

Q4. A straight conductor carries current vertically upward. Use the right-hand thumb rule to describe the direction of magnetic field on all sides. Also explain how the direction changes when current is reversed. (PYQ Pattern)

Q5. Describe the magnetic field due to a circular current-carrying loop. Draw the field pattern and explain the field at the centre of the loop. State the factors affecting field strength. (PYQ Pattern)

Q6. Explain how the magnetic field of a circular loop changes when: current is doubled, number of turns is doubled and radius is reduced. Give proper reasoning for each case. (PYQ Pattern)

Q7. What is a solenoid? Draw the magnetic field pattern of a current-carrying solenoid. Explain why the field inside is uniform and how a solenoid behaves like a bar magnet. (PYQ Pattern)

Q8. Describe how to make an electromagnet using a soft iron core. State four factors that increase its strength and mention three uses of

electromagnets. (PYQ Pattern)

Q9. Compare permanent magnet and electromagnet in detail with at least five points. Mention one use of each. (PYQ Pattern)

Q10. Explain the force on a current-carrying conductor placed in a magnetic field. State the factors affecting the force and use Fleming's left-hand rule to determine direction. (PYQ Pattern)

Q11. A conductor carrying current is placed in a uniform magnetic field. Explain what happens when: (i) conductor is parallel to field (ii) conductor is perpendicular to field (iii) current is reversed (iv) field is reversed. (PYQ Pattern)

Q12. State and explain Fleming's left-hand rule. Draw a labelled hand-rule diagram and apply it to find direction of force when current and magnetic field directions are given. (PYQ Pattern)

Q13. Describe the principle, construction and working of a simple DC electric motor. Explain the function of armature coil, split rings, brushes and magnetic field. (PYQ Pattern)

Q14. Why does the coil of a DC motor keep rotating in the same direction? Explain the role of split-ring commutator in reversing current after every half rotation. (PYQ Pattern)

Q15. Draw a labelled diagram of a simple electric motor and explain the energy conversion and practical uses of electric motors. (PYQ Pattern)

Q16. Explain electromagnetic induction with an experiment using a coil, bar magnet and galvanometer. Describe observations for motion towards, motion away and stationary magnet. (PYQ Pattern)

Q17. State Fleming's right-hand rule and explain its use in finding the direction of induced current. Compare it with Fleming's left-hand

rule. (PYQ Pattern)

Q18. Describe the principle, construction and working of an electric generator. Explain how current is induced in the coil and why the direction changes after every half rotation. (PYQ Pattern)

Q19. Draw a labelled diagram of an AC generator. Explain the roles of armature coil, magnets, slip rings and brushes. (PYQ Pattern)

Q20. Compare electric motor and electric generator in a table covering principle, energy conversion, rule used, input, output and main use. (PYQ Pattern)

Q21. Explain alternating current and direct current. Draw simple current-time graphs for AC and DC. Mention frequency of AC in India and its meaning. (PYQ Pattern)

Q22. Describe domestic electric wiring in detail. Explain live wire, neutral wire, earth wire, fuse, switch, sockets and parallel arrangement of appliances. (PYQ Pattern)

Q23. Explain why domestic appliances are connected in parallel. Include potential difference, independent control, equivalent resistance and safety of appliances. (PYQ Pattern)

Q24. What are short circuiting and overloading? Explain their causes, dangers and prevention using fuse/MCB and proper wiring. (PYQ Pattern)

Q25. Explain the importance of earthing. Describe how earthing protects a person from electric shock when the live wire touches the metallic body of an appliance. (PYQ Pattern)

Q26. Case Study: A family uses an electric iron, heater and geyser from the same socket. The fuse melts repeatedly. Answer: (i) name the fault (ii) cause (iii) protective device (iv) why fuse melts (v) two

precautions. (PYQ Pattern)